

Element B (Kamlak)

Research on western painted turtles and the red eyed slider shows that environmental limitations reduce growth and overall body condition. In captivity, western painted turtles kept in tanks that are too small or lack proper basking areas experience significant health problems. Without adequate basking, turtles cannot regulate body temperature or synthesize vitamin D3, leading to metabolic bone disease and shell infections. Additionally, restricted space increases stress and weakens the immune system, further contributing to declining health. As of right now Bella the red eyed slider Zark the western painted shell are both in the same tank. According to (Rahman, 2025) a large red eyed slider like bella is 8-12 inches. Bella is measured at 10-11 inches and he states that the minimum tank size for an adult like bella is 100 gallons and the ideal tank size is 125+ gallons with the dimensions of 72" x 18" x 24" (L x W x H) .

One scholarly study looked into the impact of environmental circumstances on turtles, including species like the Red-eared Slider and closely related painted turtles like the Western Painted Turtle, even though many controlled tests on captive turtle settings are restricted due to ethical considerations. In this study that was published on ScienceDirect, scientists set up controlled settings to see how habitat restrictions affected the growth and general health of the turtles. The results can be applied to captive habitats, such as tanks used in homes or zoos, even if the study's focus was more widely on ecological restrictions.

Turtles were put in settings with limited resources and space in this experiment. Researchers found that compared to turtles in ideal conditions, turtles in these restricted habitats had lower body masses, slower growth rates, and worse general health. This is directly related to tanks that are too small because they inhibit natural movement, swimming behavior, and exploration—all of which are critical for healthy development. An inadequately sized tank produces a constrained environment similar to the ones examined in the study, which results in behavioral and physical decline.

The basking area is another important aspect of turtle health. In order to control their body temperature and preserve the health of their shells, both bella and zark rely significantly on basking. Under ideal circumstances, turtles emerge from the water to dry off in the presence of heat and UVB rays. However, turtles cannot completely dry their shells or get the necessary UV exposure if the basking area is not ideal—that is, if it is too tiny, difficult to reach, or has inadequate illumination. Serious health issues like Metabolic Bone Disease, weakening shells, and an elevated risk of infections can result from this. Zark the smaller turtle is more affected by this because the basking area is not big enough for both of the turtles to be there and since bella is bigger and stronger she can easily just hog the basking area and there is nothing zark can do about it which is why we need to create a separate basking area for zark that only he can fit in or find a way for both bella and zark to bask together without any trouble .

Turtle well-being is significantly influenced by enrichment in addition to tank size and basking conditions. Turtles had fewer opportunities to engage with their surroundings in the study conditions due to a lack of resources. In the same way, turtles may become bored and stressed in a tank that has nothing to investigate, like rocks, plants, or structures of any kind. This lack of

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excitement may lead to decreased eating habits and activity levels, which could worsen health problems.

Regarding environmental enrichment in turtles, the National Center for Biotechnology Information study demonstrates a number of advantages and disadvantages. Enrichment (such as toys, shelters, and feeding devices) has been shown to improve overall welfare and lower stress levels while also increasing activity, decreasing boredom, and promoting natural behaviors like swimming and foraging. This strongly supports the idea that turtles thrive in environments with more engagement rather than empty tanks. However, a major drawback is that enrichment is only one aspect of good care and cannot address bad overall conditions, such as limited tank size or improper basking regions. The study also points out that it is challenging to pinpoint the precise enrichment techniques that work best because several kinds are frequently employed simultaneously, and some findings might only show short-term gains rather than long-term health consequences. Overall, even though the study unequivocally demonstrates that enrichment enhances turtle well-being, it is unable to completely replace appropriate habitat design and is restricted in its ability to isolate particular aspects.

Works cited:

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